```
    -alias ca -file "$(mkcert -CAROOT)/rootCA.pem"
```

This will ask you whether you want to trust the root CA certificate and then ask you for a password for the new keystore. Accept the certificate and type in a suitable password, then copy the generated keystore into the Natter project root directory.

Certificate chains

When configuring the trust store for your HTTPS client, you could choose to directly trust the server certificate for that server. Although this seems more secure, it means that whenever the server changes its certificate, the client would need to be updated to trust the new one. Many server certificates are valid for only 90 days. If the server is ever compromised, then the client will continue trusting the compromised certificate until it's manually updated to remove it from the trust store.

To avoid these problems, the server certificate is signed by a CA, which itself has a (self-signed) certificate. When a client connects to the server it receives the server's current certificate during the handshake. To verify this certificate is genuine, it looks up the corresponding CA certificate in the client trust store and checks that the server certificate was signed by that CA and is not expired or revoked.

In practice, the server certificate is often not signed directly by the CA. Instead, the CA signs certificates for one or more *intermediate CAs*, which then sign server certificates. The client may therefore have to verify a chain of certificates until it finds a certificate of a *root CA* that it trusts directly. Verifying a certificate chain is complex and error-prone with many subtle details so you should always use a mature library to do this.

In Java, overall TLS settings can be configured explicitly using the `javax.net.ssl.SSLParameters` class[8] (listing 7.8). First construct a new instance of the class, and then use the setter methods such as `setCipherSuites(String[])`that allows TLS versions and cipher suites. The configured parameters can then be passed when building the `HttpClient` object.